Placement for Binary-Weighted Capacitive Array in SAR ADC Using Multiple Weighting Methods

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***Abstract*—The overall accuracy and linearity of a matching-limited successive-approximation-register analog-to-digital converter are primarily determined by its digital-to-analog converter's (DAC's) matching characteristics. As the resolution of the DAC increases, it is harder to achieve accurate capacitance ratios in the layout, which are affected by systematic and random mismatches. An ideal placement for the DAC array should try to minimize the systematic mismatches, followed by the random mismatch. This paper proposes a placement strategy, which incorporates a matrix-adjustment method for the DAC, and different placement techniques and weighting methods for the placements of active and dummy unit capacitors. The resulting placement addresses both systematic and random mismatches. We consider the following four systematic mismatches such as the first-order process gradients, the second-order lithographic errors, the proximity effects, the wiring complexity, and the asymmetrical fringing parasitics. The experimental results show that the placement strategy achieves smaller capacitance ratio mismatch and shorter computational runtime than those of existing works.**

***Index Terms*—Analog placement, analog–digital-converter, capacitance mismatch ratio, capacitive array, common-centroid, digital–analog-converter, spatial correlation coefficient, successive-approximation-register.**

I. Introduction

SUCCESSIVE-APPROXIMATION-REGISTER (SAR) analog-to-digital converters (ADCs) are by far the most popular architecture for data-acquisition applications and an example of a conventional *N*-bit SAR ADC architecture is shown in Fig. 1. It consists of a sample-and-hold, a comparator, a *N*-bit capacitive digital-to-analog converter (DAC) array and a SAR control logic, where *N* is the resolution of the ADC. The ADC survey in [1] has shown that the Walden figure-of-merit (FOM) for state-of-the-art SAR ADCs is at least 10× better than other ADCs architectures and it is proven empirically that the SAR ADCs are the most energy-efficient ADCs over a broad range of resolutions [2], [3].

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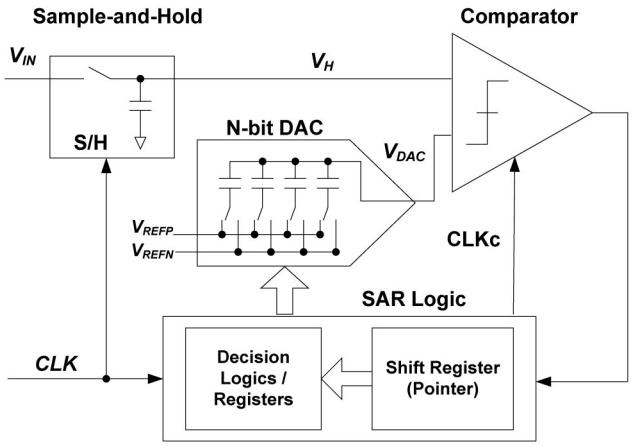


Fig. 1. Block diagram of an *N*-bit SAR ADC.

The overall accuracy and linearity of a matching limited SAR ADC are primarily determined by its capacitive DAC array matching characteristics. As the resolution of the DAC increases, it is harder to achieve accurate capacitance ratios in the layout, which are affected by the systematic and random mismatches. The systematic mismatch is caused by identical devices with asymmetrical environments, while the random mismatch is caused by statistical fluctuations in processing conditions or material properties [4], [5]. There are two popular mismatch evaluation models to quantify the random mismatch and the first-order systematic mismatch. First, the 2-D spatial correlation model helps to quantify the degree of dispersiveness in the layout [6]. A higher degree of dispersiveness indicates that there is a smaller random mismatch in the layout. Second, the 2-D oxide gradient model helps to estimate the oxide-gradient-induced mismatch (first-order systematic mismatch) in the capacitive array [7], [8].

Numerous placement methods have been reported to deal with arbitrary shapes and devices in the layout. The heuristic search approach aims to maximize the degree of dispersiveness in the layout but the placement might not have a common-centroid arrangement and thus resulting a systematic mismatch [9]. Pairs representation techniques are introduced to handle symmetry, common-centroid, and general placement simultaneously but they do not yield the optimal placement for the binary-weighted capacitive arrays [10]–[14]. The simulated-annealing-based (SA-based) approach aims to increase the degree of dispersiveness and minimize oxide-gradient-induced

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mismatch simultaneously but the resulting placement increases the complexity of the wiring [15], [16].

Besides the first-order systematic mismatch, no previous studies take into account of the following systematic mismatches for the placement of the binary-weighted capacitive array, such as the second-order lithographic errors, the proximity effects, the symmetry and the complexity of the wiring, and the far range fringing fields effects. For example, in the nanometer CMOS technologies, the interconnect parasitic capacitances have reached the order of femtofarads while the capacitance of unit capacitors in the DAC has reduced to the scale of a few femtofarads for state-of-the-art SAR ADCs [17]–[21]. Any asymmetry in the wiring scheme can easily cause more problems than are solved [5], [22]. Beside the placement constraints, many existing methods face exponential increase in the computational runtime. For example, a 10-bit capacitive array in $A_{32\times32}$ matrix has approximately 3×10^{604} number of combinations. The number of combinations for the binary-weighted capacitive arrays, grows astronomically beyond 10-bit and thus it is harder to achieve the optimization goal within acceptable computational runtime.

To deal with all these challenges in the binary-weighted capacitive array, this paper presents a placement strategy, which incorporates a matrix-adjustment method for the DAC, different placement techniques and weighting methods for the placements of active and dummy unit capacitors. This approach addresses both systematic and random mismatches and reduces the computational runtime significantly. The following systematic mismatches are considered during the placement: 1) the first-order process gradient effects; 2) the second-order lithographic errors and the proximity effects; 3) the symmetry and the complexity of the wiring; and 4) the far range fringing fields effects.

The first-order process gradient effects are minimized through the matrix-adjustment method to achieve a more symmetrical structure for any binary-weighted capacitive array. In order to alleviate the second-order lithographic errors and the proximity effects, the capacitive array is surrounded by the dummy unit capacitors to act as adjacent structures. The placement of the remaining dummy capacitors is dealt with by the spatial weighting method. This method helps to maximize the overall spatial correlation coefficient for the binary-weighted DAC capacitive array. The placement of DAC unit capacitors is dealt with by the diagonal weighting method. This method allows a dispersive placement of the DAC while maintaining symmetrical and common-centroid arrangement. The placement resembles a star-like placement and thus increases the degree of dispersiveness and simplifies the wiring complexity compared to existing works. Unequal far-range fringing fields between capacitors happen when the unit capacitors are placed in the different surroundings. This effect is amplified significantly when the physical dimensions of the unit capacitor and the minimum spacing between unit capacitors decrease in advanced CMOS technologies. This effect is minimized through the homogenization method, which addresses the existing trade-offs in [23]. The experimental results show that the placement obtained using the proposed placement strategy achieves a smaller oxide-gradient-induced mismatch and shorter computational runtime than those of existing works.

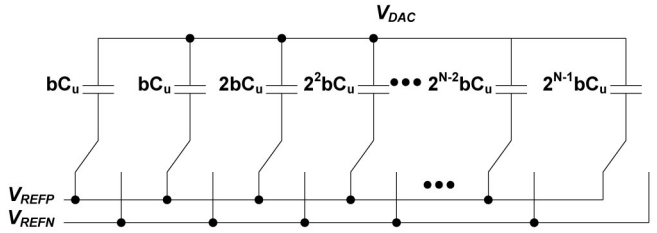


Fig. 2. N-bit binary-weighted capacitive DAC array.

The rest of this paper is organized as follows. Section II defines the terminology used in this paper and discusses the two existing evaluation models, namely the oxide gradient model [6] and the 2-D correlation model [8]. Section III discusses our proposed placement strategy. Finally, experimental results are presented in Section IV and conclusions are given in Section V.

II. Terminology and Review of Evaluation Models

A. Terms and Definitions for SAR Capacitive DAC Arrays

In a conventional N-bit binary-weighted capacitive array, as shown in Fig. 2, there are $N + 1$ capacitors denoted by $C = \{C_0, C_1, C_2, ..., C_j, ..., C_N\}$ with a ratio of $C_0 : C_1 : C_2 : C_3 :...: C_j :...: C_N$ be $b : b : 2b : 4b : ... : 2^{j-1}b : ... : 2^{N-1}b$, where $0 \le j \le N$ and $b \in \mathbb{Z}^+$. For the layout implementations, C_{typeA} in [24] and [25] and C_{typeB} in [26] and [27] are known to be the most common architectures adopted when realizing the N-bit capacitive array, where $C_{typeA} = \{C_0^A, C_1^A, C_2^A, ..., C_N^A\} = 1 : 1 : 2 : 4 : ... : 2^{N-1}$ and $C_{typeB} = \{C_0^B, C_1^B, C_2^B, ..., C_N^B\} = 2 : 2 : 4 : 8 : ... : 2^N$. C_{typeB} is often chosen in implementations because every element in C_{typeB} is made up of even number of unit capacitors and thus, helps to further improve the symmetry of the layout. C_{total} is the total number of unit capacitors in C and it is placed in a $A_{L\times W}$ matrix. The size of $A_{L\times W}$ is denoted by S, where $\{S = L\times W : L, W \in \mathbb{Z}^+\}$. Let $a_{l,w}$ denote the entry in the l-th row and w-th column, where $0 \le l \le L$ and $0 \le w \le W$. Each entry in $A_{L\times W}$ can be placed with either one unit capacitor from C or one dummy unit capacitor, C_{dummy}. The total number of dummy unit capacitors in $A_{L\times W}$ is denoted as C_D, where $C_D = S - C_{total}$.

To have a fair comparison with other previous works as those in [6], [8], [9], [15], and [16], two mismatch evaluation models are adopted in this paper [6], [8]. The following subsection reviews these two models in the context of C_{typeA} architecture and the parameters of the models are defined.

B. Random Mismatch—2-D Spatial Correlation Model

The spatial correlation model of any two unit capacitors C_{um} and C_{un}, in the C_{typeA} is defined as $\rho_{m,n}$

$$\rho_{m,n} = \rho_u^{D(m,n)} \tag{1}$$

where ρ_u is the correlation coefficient between any two unit capacitors and the distance $D(m, n)$ between C_{um} and C_{un} is

$$D_{m,n} = \sqrt{(x_m - x_n)^2 + (y_m - y_n)^2} \times L_D \tag{2}$$

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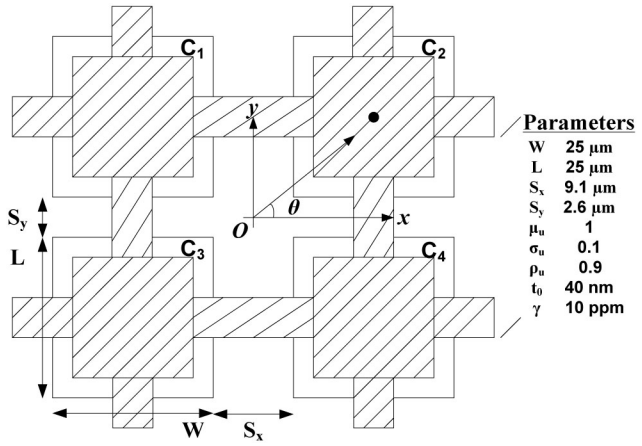


Fig. 3. Experimental setup for the geometry of unit capacitors and mismatch parameters.

where (x_m, y_m) and (x_n, y_n) are coordinates of C_{um} and C_{un}, respectively, and L_D is dependent on the process and size of the devices. To simplify (1) and (2), we assume $L_D = 1$ to observe the relationship between correlation and random mismatch. We also assume that all unit capacitances have the same variances and the correlation coefficient between any two unit capacitors ρ_u is 0.9, as shown in Fig. 3. Therefore, the variance of any capacitor in C_{typeA}, C_i^A, which consists of 2^i unit capacitors, where $0 \le i \le N-1$ can be simplified to

$$Var\left(C_i^A\right) = \sum_{a=1}^{2^i} Var(C_u) + 2\sum_{a=1}^{2^i-1}\sum_{b=a+1}^{2^i} Cov(C_{ia}, C_{ib}) = Var(C_u) \times (2^i + 2\sum_{a=1}^{2^i-1}\sum_{b=a+1}^{2^i} \rho_{a,b}) \quad (3)$$

where $\mathrm{Var}(C_u)$ is the variances of the unit capacitor and $\mathrm{Cov}(C_{ia}, C_{ib})$ is the covariance of C_{ia} and C_{ib}.

Similarly, given any two capacitors in C_{typeA}, C_i^A and C_j^A, which consist of 2^i and 2^j unit capacitors, respectively, where $0 \le i, j \le N-1$, the correlation coefficient of C_i^A and C_j^A is the summation of the correlation coefficients of all the capacitor pairs in C_i^A and C_j^A. It can be simplified to

$$\rho_{i,j} = \frac{Cov\left(C_i^A, C_j^A\right)}{\sqrt{Var\left(C_i^A\right) Var\left(C_j^A\right)}} = \frac{\sum_{a=1}^{2^i}\sum_{b=1}^{2^j} \rho_{a,b}}{\sqrt{\left(2^i + 2\sum_{a=1}^{2^i-1}\sum_{b=a+1}^{2^i} \rho_{a,b}\right)\left(2^j + 2\sum_{a=1}^{2^j-1}\sum_{b=a+1}^{2^j} \rho_{a,b}\right)}} \quad (4)$$

where $\mathrm{Cov}\left(C_i^A, C_j^A\right)$ is the covariance of C_i^A and C_j^A. $\mathrm{Var}\left(C_i^A\right)$ and $\mathrm{Var}\left(C_j^A\right)$ are variances of capacitors C_i^A and C_j^A, respectively.

For a N-bit capacitive array, the total correlation coefficient, L, is the summation of the correlation coefficients of all the capacitor pairs

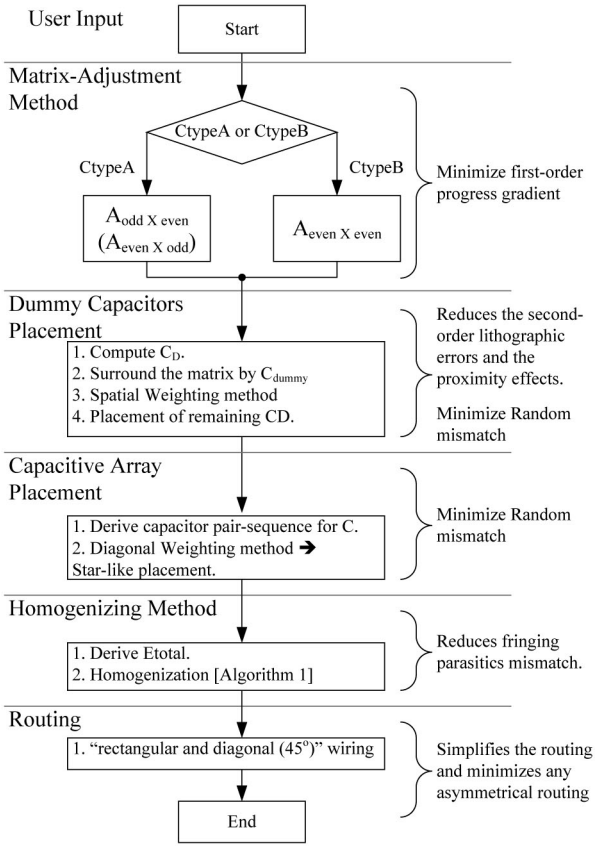


Fig. 4. Flow chart illustrating the various techniques used in the proposed placement strategy.

$$L = \sum_{i=1}^{N}\sum_{j=i+1}^{N+1} \rho_{i,j}. \quad (5)$$

It was reported that a higher total correlation coefficient, L, leads to a lower variation in the capacitive array or a higher matching capacitance ratio [6]. This model provides an insight to the degree of randomness and dispersion in the capacitive array.

C. First-Order Process Mismatch—Oxide Gradient Model

After the placement of C_{typeA}, the effective capacitor ratio becomes $C_0^{A*} : C_1^{A*} : C_2^{A*} : C_3^{A*} : \ldots : C_j^{A*} : \ldots : C_N^{A*}$. Let $\{C_{j1}^{A*}, C_{j2}^{A*}, C_{j3}^{A*}, \ldots, C_{j2^{j-1}}^{A*}\}$ denote the set of 2^{j-1} unit capacitors belonging to C_j^{A*}. The total capacitance of C_j^{A*} is the summation of the capacitance of the unit capacitors in C_j^{A*}.

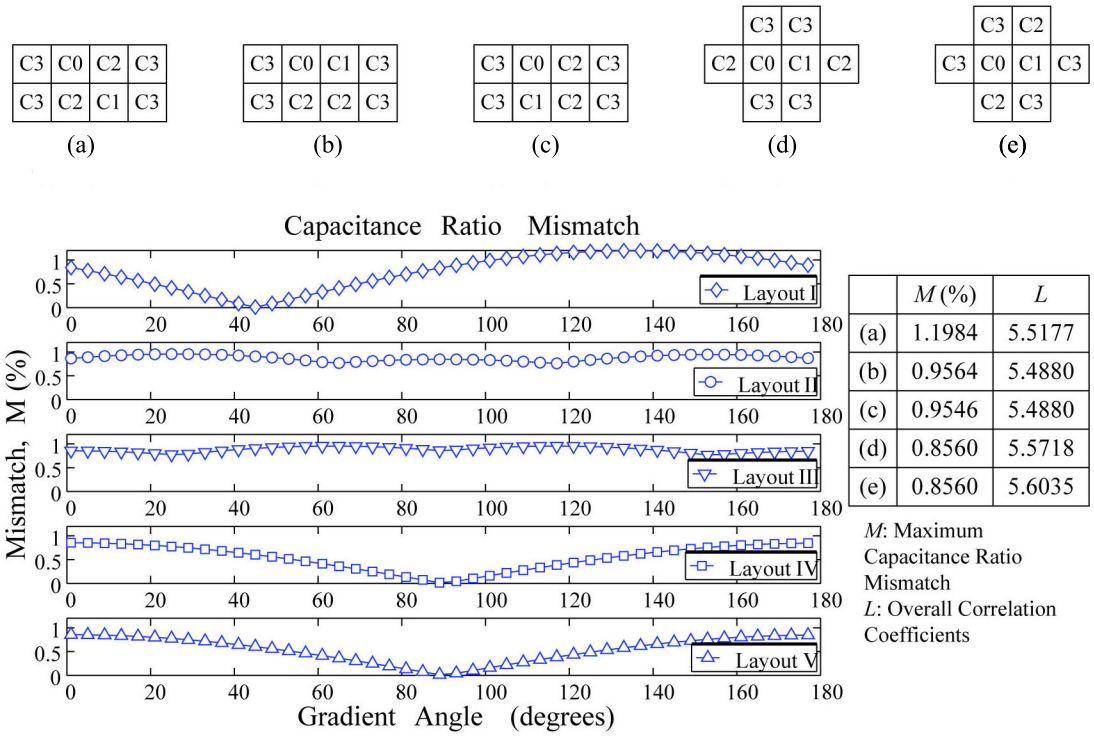
Assume that the capacitive array is affected by a linear oxide gradient γ in the direction specified by an angle θ, as shown in Fig. 3. Due to the oxide gradient, different unit capacitors located at the different locations experience different oxide thicknesses and therefore the effective capacitance of any unit capacitor is

$$C_{ji}^{A*} = C_u \frac{t_0}{t_i} \quad (6)$$

where C_u is the unit capacitance, t_0 is the oxide thickness at the origin *0* and t_i is the equivalent oxide thickness at location (x_i, y_i)

$$t_i = t_0 + \gamma(x_i \cos\theta + y_i \sin\theta). \quad (7)$$

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	M (%)	L
(a)	1.1984	5.5177
(b)	0.9564	5.4880
(c)	0.9546	5.4880
(d)	0.8560	5.5718
(e)	0.8560	5.6035

Fig. 5. Experimental placements for 3-bit C_{typeA} in $A_{2\times4}$ and $A_{3\times4}$. (a) Layout I. (b) Layout II. (c) Layout III. (d) Layout IV. (e) Layout V.

The ratio mismatch is then calculated for all the pairs of capacitor in C_{typeA}. The capacitance ratio mismatch, M in C_{typeA} is defined as

$$M = \max\left(\left|\frac{\frac{C_j^{A*}}{C_k^{A*}} - \frac{C_j^A}{C_k^A}}{\frac{C_j^A}{C_k^A}}\right|\right) \times 100\% = \max\left(\left|\frac{C_j^{A*}}{C_k^{A*}}\frac{C_k^A}{C_j^A} - 1\right|\right) \times 100\% \quad (8)$$

where $0 \le j, k \le N+1$. Since the gradient angle cannot be predicted, the mismatch is estimated through $0^\circ \le \theta \le 180^\circ$ and the maximum mismatch value is obtained.

This model provides an insight to the degree of symmetry and coincidence which is very important for binary-weighted capacitive array in nanometer CMOS technologies [7]. A smaller capacitance mismatch ratio, M, yields a smaller systematic mismatch in the capacitive array. To have a fair comparison with other works, we assume that the oxide thickness t_0 is 40 nm, the oxide gradient γ is 10 parts per million and the width and length of a unit capacitor, C_u is 25 μm (W) by 25 μm (H) with a spacing of 9.1 μm (S_x) by 2.6 μm (S_y), respectively, as shown in Fig. 3.

III. Placement Strategy

The proposed placement strategy first adjusts the matrix size based on the parameters listed in Fig. 3 and the type of binary-weighted capacitive array under consideration, e.g., C_{typeA} or C_{typeB}. As it will be explained in Section III-A, the matrix size has a great influence on the final placement result and the degree of symmetry (first-order process gradient effects). Our next step is to prioritize of the placement of the dummy capacitors and the capacitive array, respectively. This eliminates the second-order lithographic errors and the proximity effects and maximizes the degree of dispersiveness that can be achieved by the capacitive array. In order to simplify the routing in the common-centroid layout, the diagonal weighting is proposed for the placement of the capacitive array and thus allowing us to use a simple rectangular and diagonal wiring method to connect the capacitors together. Finally, the homogenizing algorithm is applied to the matrix and this allows a better distribution of parasitic capacitances amongst the capacitive array. The flow chart, as shown in Fig. 4, provides the details of our proposed placement strategy.

A. Matrix-Adjustment Method

Based on the oxide gradient model and the spatial correlation model, we investigate all the different possible layouts for the 3-to-5-bit C_{typeA} and C_{typeB} capacitive arrays in different matrix sizes. Five different examples of a 3-bit C_{typeA} layout are presented in Fig. 5, for the discussion of our matrix-adjustment method. Fig. 5(a) shows a layout (Layout I) in $A_{2\times4}$ with partial symmetry about the origin where C_0^A and C_1^A are placed diagonally. Fig. 5(b) and (c) shows layouts (Layout II and III) in $A_{2\times4}$ with partial symmetry about the y-axis and the x-axis, respectively, where C_0^A and C_1^A are placed on adjacent sides. This adjacent side placement of C_0^A and C_1^A resulted in decreased M and L, as shown in the table in Fig. 5. Fig. 5(d) and (e) show layouts (Layout IV and V) in $A_{3\times4}$ with symmetry about the x-axis and partial symmetry about the origin, respectively. By changing the matrix to $A_{3\times4}$, there are further improvements in M and L. The results were calculated based on the input parameters listed in Section II except that

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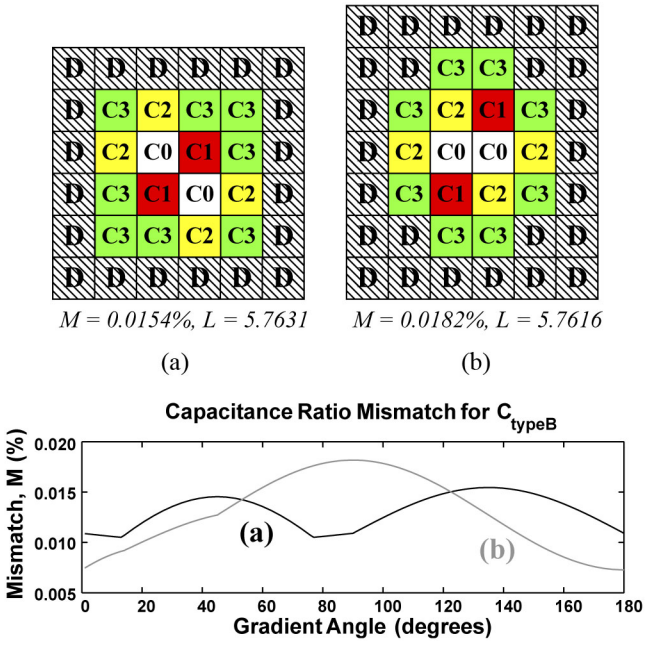


Fig. 6. Experimental placements for 3-bits C_{typeB} capacitive arrays in $A_{6\times6}$ and $A_{7\times6}$. (a) Layout V in $A_{6\times6}$. (b) Layout V in $A_{7\times6}$.

S_x and S_y are 9.1 μm as this allows equal spacing between unit capacitors. The plots of capacitance ratio mismatch variation with respect to the gradient angle for these placements, as shown in Fig. 5, provide us with insightful details about the symmetrical property of the layout. Based on all the layouts investigated in this experiment, the following observations were made.

1) C_{typeA} contains an odd-numbered unit capacitor in C_0^A and C_1^A and thus, it is harder to achieve minimal M in $A_{odd\times odd}$ or $A_{even\times even}$. $A_{odd\times even}$ or $A_{even\times odd}$ provides a better symmetry for C_{typeA} capacitive arrays.
2) Adjacent side placements for the odd-numbered capacitors yield a smaller M. Thus, it is preferred that C_0^A and C_1^A are placed on adjacent sides and they should be placed near the center of a matrix.
3) The diagonal placements of capacitor pairs, from C_2 to C_N, yield a higher L than adjacent side placements.
4) C_{typeA} achieves better symmetry when it has a symmetrical mismatch variation.

The same evaluation method is applied to C_{typeB} and Fig. 6 shows an example of two different layouts of a 3-bit C_{typeB} in $A_{6\times6}$ and $A_{7\times6}$. The diagonal placements of the even-numbered capacitor pairs and the matrix-adjustment to $A_{even\times even}$ in the C_{typeB} has reduced M and improved L. Therefore, we propose to adjust the size of the matrix, $A_{L\times W}$ according to the types of capacitive array. The modification to the matrix size does not have a significant impact on the overall area.

B. Dummy Capacitors Placement Using Spatial Weighting Method

After the matrix has been adjusted to achieve minimal M, the next priority is the placement for the dummy unit capacitors. Although the dummy unit capacitors do not induce mismatches, their location affects the matching of the capacitive array. To minimize these mismatches, it is important to consider the placement for the dummy unit capacitors. C_D is computed based on the matrix size and the capacitive array. These dummy unit capacitors are first added into the perimeter of the capacitive array to act as adjacent structures and therefore minimizes the second-order lithographic errors. Two weighting methods are evaluated to determine the best choice for prioritizing the location for the remaining dummy unit capacitors in $A_{L\times W}$.

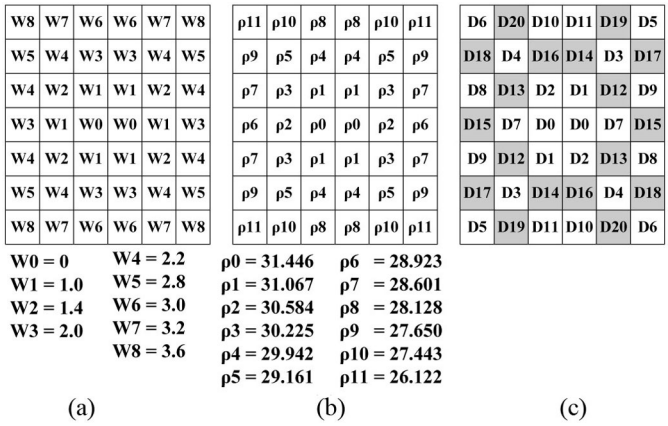


Fig. 7. Illustrations of entry weight definitions. (a) Distance weight. (b) Spatial correlation weight. (c) Diagonal weight.

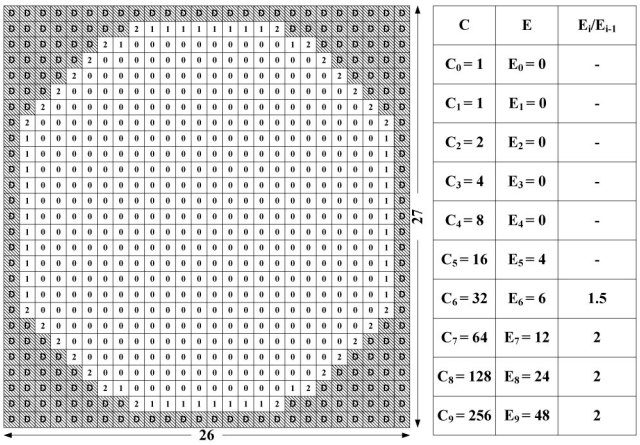
The distance weighting method was first used for the matrix to pair-sequence transformation [8], [16]. The distance from the entry $a_{l,w}$ to the center of $A_{L\times W}$ is defined as

$$d_{l,w} = \sqrt{(x_l - x_c)^2 + (y_w - y_c)^2} \tag{9}$$

where $(x_l,\ y_w)$ is the coordinate of $a_{l,w}$ and $(x_c,\ y_c)$ is the coordinate of the center entry of $A_{L\times W}$. The entries with the same distance are classified into the same weight and these weights are prioritized from the shortest distance to the longest distance.

The spatial weighting method is based on the spatial correlation model [9]. $L_{l,w}$ denotes the sum of the correlation coefficients between the entry $a_{l,w}$ and the rest of dummy unit capacitors. Due to the symmetry of the matrix, the entries with the same total correlation coefficient are classified into the same weight and these weights are prioritized from the highest total spatial correlation coefficient to the lowest total spatial correlation coefficient. Fig. 7(a) and (b) show examples of the completed entry assignment using the distance weighting method and the spatial weighting method for the matrix $A_{7\times6}$, respectively. Note that $W0$ and $\rho0$ denote the shortest $d_{l,w}$ and highest $L_{l,w}$, respectively. Comparing the distance weighting method and the spatial weighting method, the latter method provides a clearer differentiation between the different types of entries. Therefore, the spatial weighting method is used for the placement of the dummy unit capacitors. The dummy unit capacitors are assigned starting from the smallest $L_{l,w}$ so that the capacitive array is able to achieve the maximal L in $A_{L\times W}$. A placement layout of the dummy capacitors for a 9-bit C_{typeA} capacitive array is shown in Fig. 8.

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C	E	E_i/E_{i-1}
$C_0 = 1$	$E_0 = 0$	-
$C_1 = 1$	$E_1 = 0$	-
$C_2 = 2$	$E_2 = 0$	-
$C_3 = 4$	$E_3 = 0$	-
$C_4 = 8$	$E_4 = 0$	-
$C_5 = 16$	$E_5 = 4$	-
$C_6 = 32$	$E_6 = 6$	1.5
$C_7 = 64$	$E_7 = 12$	2
$C_8 = 128$	$E_8 = 24$	2
$C_9 = 256$	$E_9 = 48$	2

Fig. 8. Placement dummy unit capacitors and dummy edge computation for 9-bit C_{typeA} capacitive array with dummy unit capacitors in $A_{27\times26}$.

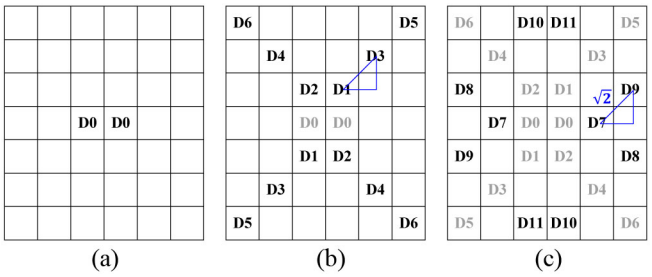


Fig. 9. Diagonal weighting procedure. (a) Location of first entry pair, D0. (b) Location of entry pairs, D1 to D6. (c) Location of entry pairs, D7 to D11.

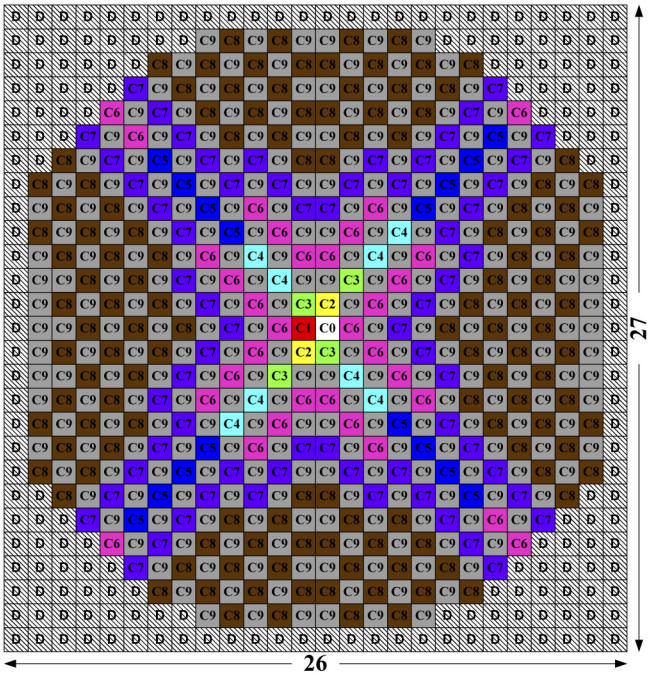


Fig. 10. Placement result of 9-bit C_{typeA} capacitive array with dummy unit capacitors in $A_{27\times26}$.

C. Capacitive Array Placement Using Diagonal Weighting Method

The next priority is the placement of capacitive array, C. According to the experiment in Section III-A and the spatial weighting method, we proposed a diagonal weighting method for the placement of capacitive array.

In the diagonal weighting method, a pair of two diagonal entry weights is defined to be diagonally opposite and equidistant with respect to the center of $A_{L\times W}$. This method is broken down into the following steps.

1) The first pair of entry weights in C_{typeA}, denoted as $D0$, is the first two entries closest to the center of the matrix, $A_{odd\times even}$. Fig. 9(a) illustrates the first pair of entry weights, $D0$ in $A_{7\times6}$.
2) The next entry pair, $D1$, is placed one unit distance away from the previous pair, $D0$, starting clockwise from the 12-o'clock position. $D2$ is then placed mirroring $D1$ about the center of the matrix.
3) Step 2 is repeated iteratively where the subsequent odd-entry pairs are placed in clockwise order from the 12-o'clock position at $\sqrt{2}$ unit distance away from the previous pair of odd-entry diagonal weights. The following even-entry pair is placed mirroring the previous odd-entry pair. This sequence repeats till the entry pairs reach the boundary of the dummy unit capacitors. Fig. 9(b) illustrates the location of entry pairs, $D1$ to $D6$ in $A_{7\times6}$. It is noted that the elements placed resemble the letter X.

Algorithm 1 Homogenization

Input: C, E, $A_{L\times W}$
Output: $A_{L\times W}$

```
while E ≠ E_c do
    for i = 0 to N do
        while E[i] > E_c[i] do
            Search for highest diagonal entry weight ∈ C_i;
            Search for lowest diagonal entry weight ∈ C_j
            whose E[j] < E_c[j] and j > i;
            Swap the entry pair-sequence and update E_c
        end while
    end for
end while
```

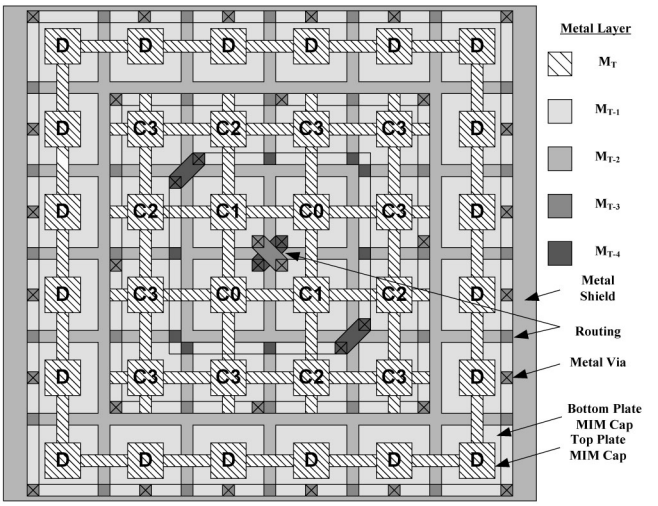


Fig. 11. Layout details showing the parasitic shielding and 45° routing used in 3-bits C_{typeB} capacitive array layout.

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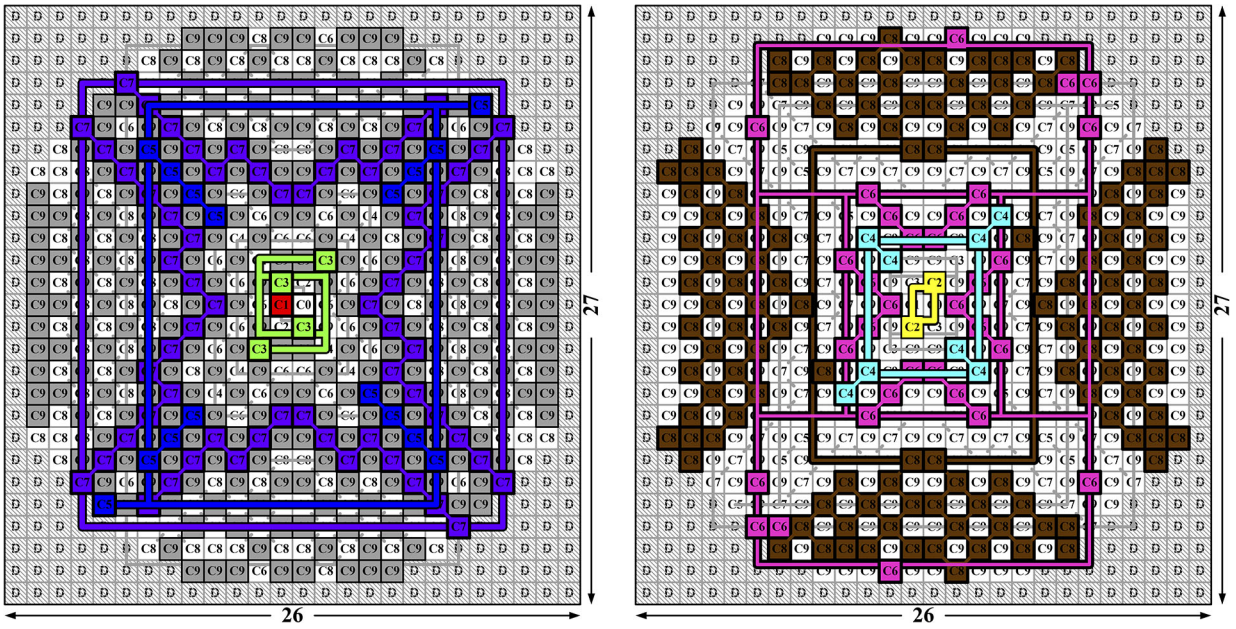


Fig. 12. Final placement result of 9-bit C_{typeA} capacitive array with dummy unit capacitors in $A_{27\times26}$. The colored rectangles and diagonal wires represent the routing connection of the capacitors.

4) The next pair of entry weights, $D7$, is placed to the nearest empty entry clockwise from the 3-o'clock position with respect to the center of the matrix.
5) The following even-entry pair, $D8$, is placed clockwise from the 3-o'clock position $\sqrt{(2i+k_x)^2+k_y^2}$ unit distance away from the center of the matrix, where i increments from 1 and (k_x, k_y) is equivalent to (0.5, 1).
6) The subsequent even-entry pair, $D10$, is then placed diagonally opposite to $D8$ with respect to the diagonal line forming the bottom right leg of the "X." All subsequent even-entry pairs are then placed sequentially above and below this diagonal line until they reach the boundary of the dummy unit capacitors.
7) The corresponding odd-entry pairs are placed mirroring the even-entry pairs about the y-axis.
8) Steps 4 to 7 are repeated iteratively until Step 4 reaches the boundary of the dummy unit capacitors.
9) The remaining diagonal entry weights are defined using the distance weighting method [16]. Fig. 7(c) shows an example of completed assignment with diagonal weighting method for matrix $A_{7\times6}$.

The diagonal weighting method for the C_{typeB} is similar to C_{typeA} except that the first two pairs of entry weights, $D0$ and $D1$, are selected from the 12-o'clock and 3-o'clock positions at the closest distance to the center of the matrix $A_{even\times even}$ and (k_x, k_y) is equivalent to (1.5, 0.5).

A capacitor pair-sequence is formed by iteratively pairing two unit capacitors in C, starting from C_0 to C_N, which is modified from [16]. Similarly, a diagonal pair-sequence is formed starting from $D0$. Then, the capacitor pairs are assigned to the corresponding diagonal pairs. Fig. 10 shows the placement of C_{typeA} in $A_{L\times W}$.

Our proposed placement method for capacitive array ensures that unit capacitors in C_N are placed adjacent sides to the rest of unit capacitors and allows C_N to be evenly distributed across the matrix. Since the majority of unit capacitors from C_2 to C_{N-1} are surrounded by C_N, electrical mismatch is minimized. We assume that there are negligible far range fringing effects from capacitors diagonal to the unit capacitor. Compared to Hakkarainen and Nurminen's method [23], the proposed method does not require additional dummy capacitors to improve the electrical matching.

D. Homogenizing Algorithm With Edge Weighting Method

To further improve the electrical mismatch for unit capacitors adjacent to the dummy capacitors, we focus on the optimization of the placements of these unit capacitors such that the fringing field effects are distributed across the capacitive array in a binary fashion.

Any entry, $a_{l,w}$ with an active unit capacitor, is assigned with an edge weight, which is denoted by $E_{l,w}$, where $E_{l,w}$ is the sum of dummy unit capacitors that are placed adjacent side to the entry $a_{l,w}$. Let E_c denote a set of edge weights belonging to the capacitive array with a ratio $E_{c,0} : E_{c,1} : E_{c,2} : \ldots E_{c,j} : \ldots : E_{c,N}$, where $E_{c,j}$ is the summation of all $E_{l,w}$ belonging to C_j. Let E denote a set of edge weights with a ratio $E_0 : E_1 : E_2 : \ldots E_j : \ldots : E_N$, where E_j is defined as

$$E_j = \frac{E_{total}}{2^{j+1}} + \frac{2^{j+1} + 2 - mod(E_{total} - (2^{j+1} - 2), 2^{j+1})}{2^{j+1}} \tag{10}$$

and E_{total} denotes the sum of all the edge weights in $A_{L\times W}$. E_j provides an estimated number of unit capacitors from C_j that is required to be placed adjacent side to dummy unit capacitors. The goal is to ensure that the ratio between E_j and E_{j-1} is close

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TABLE I
EXPERIMENTAL RESULTS WITH OUR PROPOSED PLACEMENT AND COMMON-CENTROID PLACEMENT IN [14]. (M: CAPACITANCE MISMATCH RATIO; L: OVERALL CORRELATION COEFFICIENTS)

C_{typeA} Capacitive Array									
Array Name	Matrix Size	Method 1 [14]		Method 2 [14]		Matrix Size	Our Work		Time (s)
		M^1	L^2	M^1	L^2		M^1	L^2	
SAR_2bit	2 × 2	1.103	2.727	1.102	2.762	5 × 4	0.692	2.762	0.1
SAR_3bit	2 × 4	2.264	5.203	1.103	5.502	5 × 6	0.692	5.604	0.1
SAR_4bit	4 × 4	3.345	8.556	1.103	9.263	7 × 6	0.692	9.326	0.1
SAR_5bit	4 × 8	5.619	12.113	1.103	13.707	9 × 8	0.692	13.978	0.1
SAR_6bit	8 × 8	7.984	16.381	1.103	19.128	11 × 10	0.692	19.408	0.1
SAR_7bit	8 × 16	12.706	19.898	1.103	24.653	15 × 14	0.692	25.331	0.2
SAR_8bit	16 × 16	17.926	23.729	1.103	30.913	19 × 18	0.692	31.692	0.3
SAR_9bit	16 × 32	28.468	25.363	1.120	35.381	27 × 26	0.692	37.875	1.0
SAR_10bit	32 × 32	40.963	27.406	1.539	41.228	35 × 34	1.030	42.554	11.2
SAR_11bit	32 × 64	68.226	26.913	3.214	42.477	49 × 48	1.400	47.012	35
SAR_12bit	64 × 64	105.565	27.990	4.786	45.705	67 × 66	3.899	48.589	440
C_{typeB} Capacitive Array									
Array Name	Matrix Size	Method 1 [14]		Method 2 [14]		Matrix Size	Our Work		Time (s)
		M^1	L^2	M^1	L^2		M^1	L^2	
SAR_2bit	2 × 4	0.010	2.865	0.010	2.865	6 × 4	0.015	2.872	0.1
SAR_3bit	4 × 4	0.025	5.651	0.015	5.740	6 × 6	0.017	5.765	0.1
SAR_4bit	4 × 8	0.068	9.163	0.044	9.409	8 × 8	0.026	9.550	0.1
SAR_5bit	8 × 8	0.133	13.283	0.066	14.009	10 × 10	0.066	14.135	0.1
SAR_6bit	8 × 16	0.336	17.281	0.178	18.843	14 × 14	0.131	19.383	0.2
SAR_7bit	16 × 16	0.619	21.257	0.267	24.580	18 × 18	0.269	24.746	0.3
SAR_8bit	16 × 32	1.492	23.385	0.712	24.580	26 × 26	0.358	30.861	0.9
SAR_9bit	32 × 32	2.719	25.537	1.085	34.401	34 × 34	1.085	34.774	10.4
SAR_10bit	32 × 64	6.572	25.296	2.855	36.188	48 × 48	1.354	39.880	32
SAR_11bit	64 × 64	12.402	26.394	4.367	39.258	66 × 66	4.022	40.125	400

[1]Assumes that the oxide thickness, t_0, is 40 nm, the oxide gradient, γ, is 10 parts per million and the width and length of a unit capacitor, C_u, is 25 μm (W) by 25 μm (H) with a spacing of 9.1 μm (S_x) by 2.6 μm (S_y) respectively.
[2]Assume that all unit capacitances have the same variances and the correlation coefficient between any two unit capacitors ρ_u is 0.9.

to 2 and $E_{c,j}$ is equivalent to E_j. This method allows a better distribution of fringing fields amongst the capacitive array and improves the overall electrical matching. Fig. 8 provides an example of dummy edge computation for a 9-bit C_{typeA} capacitive array.

After an initial placement has been generated, a homogenizing algorithm is initialized and $E_{l,w}$ entries, which are greater than zero, are swapped to meet the goal (the pseudocode is shown in Algorithm 1). We adopt a look-up table to speed up the swapping process and update the pair-sequence order.

E. Routing

The star-like placement technique allows the capacitors C_N and C_{N-1} to be distributed evenly and diagonally across the layout. This placement allows rectangular and diagonal (45°) wiring technique to be used so that a simple and symmetrical routing can be achieved. Each capacitor would only require one to two rectangular wires and several diagonal (45°) wires for routing. To avoid any congestion, the rectangular wires and the diagonal wires can be drawn on different metal layers. Furthermore, the DAC switches, as shown in Fig. 2, can be sub-divided into four smaller sets of switches and placed at different side of the array so that the long rectangular wires can be totally removed. Further protection against routing parasitics is provided by a grounded metal shield over the entire array with one hole per unit capacitor to access each bottom plate [28]. While this method adds extra parasitic capacitance from both the bottom plate and the top common plate to ground, the shielding makes the weighted capacitors insensitive to other routing parasitics. Fig. 11 illustrates an example

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TABLE II
STUDY OF CAPACITANCE MISMATCH RATIO, M, ON 8-TO-11-BIT C_{typeA} WITH VARYING OXIDE THICKNESS, t_0 AND OXIDE GRADIENT, γ

Oxide thickness, t_0		30	40	50	60
Name	Size	Capacitance mismatch ratio, M			
SAR_8bit	19 × 18	1.034	0.774	0.619	0.515
SAR_9bit	27 × 26	1.049	0.774	0.619	0.515
SAR_10bit	35 × 34	1.761	1.084	0.755	0.515
SAR_11bit	49 × 48	2.445	1.465	0.998	0.735
Oxide gradient, γ		5	10	15	20
Name	Size	Capacitance mismatch ratio, M			
SAR_8bit	19 × 18	0.386	0.774	1.163	1.554
SAR_9bit	27 × 26	0.386	0.774	1.256	1.978
SAR_10bit	35 × 34	0.386	1.084	2.164	3.655
SAR_11bit	49 × 48	0.386	1.465	3.032	5.201

[1]Assumes that the width and length of a unit capacitor, C_u, is 25 μm (W) by 25 μm (H) with a spacing of 5.3 μm (S_x) by 5.3 μm (S_y) respectively.

of a 3-bit C_{typeB} capacitive array, which was built by connecting metal-insulator-metal (MIM) capacitors with routing below the shielding layer.

F. Example of 9-bit C_{typeA}

An example of a capacitor placement for 9-bit C_{typeA} with dummy unit capacitors in $A_{27\times26}$ is shown in Fig. 12. The capacitive array is surrounded by the dummy unit capacitors to minimize the second-order lithographic errors and the proximity effects. The dummy unit capacitors are placed strategically to maximize the overall spatial correlation coefficient for the capacitive array. The placement resembles a star-like pattern and thus maximizing the overall spatial correlation coefficient as well as to achieve a simple and symmetrical wiring connection. The colored rectangular wires and diagonal wires represent the wiring of the capacitors. The wiring scheme shows that the connections are much simpler and symmetrical compared to the routing methods presented in [8], [22], and [16]. The adjustment of the unit capacitors from C_5^A to C_9^A helps to achieve a more homogeneous distribution of fringing fields at the edge of the capacitive array within C_{typeA}. This helps to minimize the mismatch due to the unequal fringing mismatch.

IV. EXPERIMENTAL RESULTS

The proposed placement algorithm for the conventional binary-weighted capacitive-array DAC was implemented in the MATLAB programming language and simulated on a 3.00-GHz Intel Core2 Duo computer with 4.0 GB-RAM.

In this experiment, the proposed placement has been implemented on the 2-to-12-bit C_{typeA} and the 2-to-11-bit C_{typeB} arrays. The capacitive arrays with nonsquare placements were placed in a square matrix with the use of additional dummy

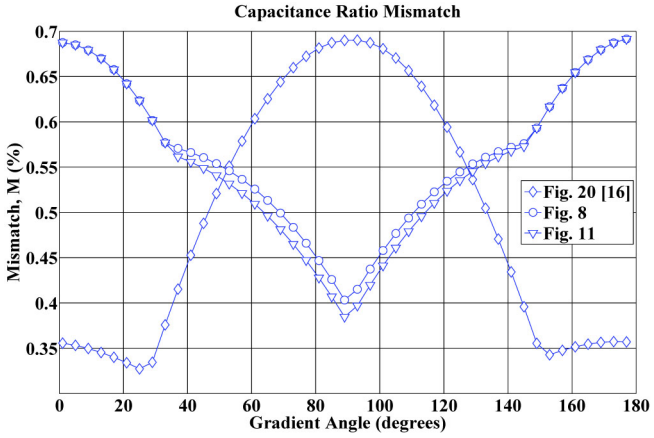


Fig. 13. Oxide-gradient-induced mismatches of three different placements for 9-bit C_{typeA} capacitive array.

capacitors so that it can achieve better matching than a rectangular matrix. For each type of the capacitive arrays, the value of the capacitance mismatch ratio, M, the value of the overall correlation coefficient, L, and the computational runtime are listed. The runtime is acquired based on an average of 100 iterations. Table I shows the experimental results for the 2-to-12-bit C_{typeA} and the 2-to-11-bit C_{typeB} capacitive arrays.

From our experimental results in Table I, the C_{typeB} capacitive array achieves a smaller M than the C_{typeA} capacitive array for resolutions below 8 bit. M begins to increase for both types of capacitive array when the resolution increases beyond 9-bit resolutions. As seen in (7) under the Section II-C, the parameters, t_0, γ, S_x, S_y, W, and L affect the effective capacitance of the unit capacitor due to the variation of the oxide thickness at different locations. Therefore, the second term in (7) has a greater influence on the variation of the effective oxide thickness when the size of the matrix increases. From our experimental results in Table II, M is minimized through increasing oxide thickness, t_0, or decreasing oxide gradient, γ, for higher resolution DACs. However, t_0 and γ are process dependent parameters, thus the size of unit capacitor (W and H) and the spacing between unit capacitors (S_x and S_y) can only be adjusted to minimize M for the higher resolution DACs. Our proposed routing method allows the spacing between the unit capacitors to be minimized, which also minimizes M. Comparing the experimental results (Table I) for the layout using the methods in [13] and [14], which are one of the common-centroid layouts, our proposed placement has increased the degree of dispersiveness and reduced the capacitance mismatch ratio for both types of capacitive arrays.

Table III compares our works with other state-of-the-art works [9], [16]. Fig. 13 shows the mismatch variation with the gradient angle for the three different placements of the SAR_9bit. From these results, our proposed placement strategy achieves a smaller capacitance mismatch ratio and a faster computational runtime with a comparable overall spatial correlation coefficient.

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TABLE III
COMPARISONS OF CAPACITANCE MISMATCH RATIO, OVERALL CORRELATION COEFFICIENTS, AND RUNNING TIME FOR THE HEURISTIC ALGORITHM [9], SA ALGORITHM [16], AND OUR WORK. (M: CAPACITANCE MISMATCH RATIO; L: OVERALL CORRELATION COEFFICIENTS)

Array Name	Matrix Size	Heuristic Algorithm [9]			SA Algorithm [16]			Matrix Size	Our Work		
		M^1	L^2	Time (s)	M^1	L^2	Time (s)		M^1	L^2	Time (s)
SAR_8bit	16 × 16	0.800	32.074	602	0.695	32.111	235	19 × 18	0.692	31.692	0.3
SAR_9bit	32 × 16	1.077	38.072	20503	0.878	38.654	681	35 × 18	0.856	36.625	1.0
SAR_9bit	23 × 23	-	-	-[3]	0.695	39.243	820	27 × 26	0.692	37.875	1.0
SAR_10bit	32 × 32	-	-	-[3]	1.146	45.515	5130	35 × 34	1.030	42.554	11.2

[1]Assumes that the oxide thickness t_0 is 40 nm, the oxide gradient γ is 10 parts per million and the width and length of a unit capacitor, C_u is 25 μm (W) by 25 μm (H) with a spacing of 9.1 μm (S_x) by 2.6 μm (S_y) respectively.
[2]Assume that all unit capacitances have the same variances and the correlation coefficient between any two unit capacitors ρ_u is 0.9.
[3]No result is reported

V. CONCLUSION

In conclusion, it is important to achieve a zero mean offset or zero systematic mismatch in the binary-weighted capacitive array as random mismatch cannot be totally eliminated. The proposed placement strategy considers the random mismatch and the four systematic mismatches (such as the first-order process gradient, the second-order lithographic errors, the proximity effects, the wiring complexity, and the far range fringing fields effects) during the placement of the capacitive array. This results in a more symmetrical and common-centroid placement for the capacitive array, which is evident from the reduction of first-order oxide gradient. Additional dummy unit capacitors are placed adjacent to the capacitive array to reduce the second-order lithographic errors and the proximity effects. The capacitive array resembles a star-like placement so that it simplifies the wiring scheme and minimizes the number of asymmetrical wiring compared to existing works. A homogenization algorithm is proposed to equalize fringing mismatches among the capacitive array. Our analytical approach has improved the computational runtime significantly while achieving comparable degree of dispersiveness compared to the state-of-the-art works.

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